Dae Seok Song

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AI Engineer | Medical AI and Modular Neural Networks

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Research Profile

AI engineer who has led model development, compression, clinical-product integration, and international regulatory evidence generation at a medical-AI startup since November 2022. My research asks whether a functional module keeps its role when moved into a separately trained model (so far tested only across models that share architecture and either differ in initialization method or share initial weights and differ in the seed-dependent train/validation split), and tests this with intervention: swapping a module between separately trained models and measuring what breaks (preliminary; module removal was recorded only as a preliminary metric, not a result).

Research Interests

Functional-module transplant and intervention; orthogonal alignment of module outputs; modular and continual learning; model compression and structured pruning; weight-transport-free learning rules.

Independent Research

Modular Neural Network: Detachable Functional Modules and Routing 2026–Present
Independent Researcher Seoul, Korea

- Designed a modular architecture with interchangeable slots, detachable functional modules, and a router.
- Across three completed variants of a two-module image classifier (hierarchical readout without a router; a router trained from the start; a router with an added "unknown" option), an orthogonal Procrustes alignment (no retraining) recovered 99.4–99.8% of the performance lost under naive swaps, covering every directed donor-to-recipient pair among four separately trained models per variant (CIFAR-10, single seed 42; run-to-run variation not measured). The models shared the dataset, architecture, and role assignment (roles were assigned by me, not learned) but differed in initialization method. Preliminary, unpublished; prior work on model stitching and re-basin already aligns independently trained networks, so the recovery is not claimed as new.
- Long-term interest: module-level control of AI behavior, i.e., removing parts that could harm people. Module removal was recorded only as a preliminary metric in the same line of experiments, and I have not measured whether removal suppresses an unwanted behavior, so current results cover only module swapping and alignment. Prior work on module-level intervention exists (e.g., Gradient Routing, SteerMoE); defining and separating harm is an open problem, and the removal comparison is planned.

Expert Synapse: Task Heads on Frozen Pretrained Backbones (design stage) 2026–Present
Independent Researcher Seoul, Korea

- Designing, with no results yet, task heads that attach to an unseen pretrained vision backbone without retraining: the backbone stays frozen, expert bundles and anchor-based signatures link the head to it, and four controls (including a learned adapter and a closed-form orthogonal Procrustes translation) are fixed in advance. Stage-0 checks are under way; main runs have not started. I claim no novelty over relative representations (Moschella et al., 2023) or Procrustes translation (Maiorca et al., 2023), and no emergent modularity.

Weight-Transport-Free vs. Backpropagation Learning-Rule Comparison 2026
Independent Researcher Seoul, Korea

- Compared a weight-transport-free learning rule (error returned through a separately learned feedback pathway instead of the transposed forward weights) against backpropagation in a preliminary comparison.
- Backpropagation led in this preliminary comparison (three-seed average accuracy 90.8% vs. 77.5%; the settings differ in one convolution option and the tuning budget was not matched, so I cannot say whether the gap is due to the rule or the setup).
- Applied Procrustes alignment to modules swapped between models trained with this rule that shared initial weights and differed in the seed-dependent train/validation split (pairs with different initialization are untested): accuracy fell on average to roughly half under naive swaps and recovered to a mean of 76.0% (77.5% before swapping; three seeds, preliminary); I then paused this line to focus on intervention-based measurement and plan to resume it under supervision.

Professional Experience

Connecteve Co., Ltd. Nov 2022–Present
AI Engineer / Early Team Member Seoul, Korea

- Consolidated 10 production knee X-ray models into 2 uncertainty-aware, partial-label multi-task models and added a keypoint model (3 models in total). Against the current product engine on the same CPU server and bilateral films, stored weights are 10.97× smaller (multi-task consolidation plus a tiny backbone) and wall-clock inference time per image is 1.74× shorter; the overall F1 on evaluation sets from three external institutions did not drop (0.016 higher; single seed).

- Rebuilt the optimization-to-serving line (ONNX optimization, 16-bit inference, BentoML serving): engine size 47% smaller and inference 3× faster than the same engine before optimization, with a 0.06-percentage-point accuracy drop on clinical comparison data. These were measured on the earlier 10-model engine, so they are reported separately from the consolidation figures above.
- Tuned per-grade decision thresholds with Youden's J statistic across four tie-break strategies, reaching or exceeding the F1 of an FDA-cleared competitor product on a matched evaluation.
- Built the standard PyTorch Lightning + Hydra experiment pipeline for medical foundation models (DINOv3, Rad-DINO) with selectable linear probing or LoRA.
- Built an in-house MLOps console (GUI/CLI) spanning training, optimization, packaging, and serving, with a two-stage hyperparameter search (LLM-agent proposals, then Optuna/TPE after ten stalled rounds); in a from-scratch knee-detection run, switching to Optuna raised mAP by 6.7 points (0.8022 → 0.8693). Evaluated 12 agentic-workflow frameworks and selected LangGraph; reworked the deployment pipeline into an agentic workflow on vLLM (Qwen, Mistral) as a proof of concept; caching cut re-onboarding from 3 h 36 min (cold) to 38 min (warm), 5.7×.
- Led authoring of the AI technical documentation for CONNEVO KOA and took part in the full approval process: cleared in Korea, approved in the UAE, CE MDR certified, and granted by Singapore HSA; FDA process in progress.

Minerva Soft Mar 2022–Jul 2022
Image Processing Intern Korea

- Replaced full-image barcode scanning with an OpenCV ROI pipeline, raising recognition from 70% to 90%; shipped a C++ DLL after 170,000+ memory-leak and stability iterations.

Education

University of Suwon Mar 2017–Feb 2023
B.S. in Information Security, Valedictorian GPA 4.29/4.50; Major 4.37/4.50

Selected Publications

Hwang, Y., Lee, Y., **Song, D. S.**, Kang, D.-H., Ro, D. H. "Beyond Landmarks: Anchor-Guided Orientation Regression for Robust Spinopelvic Measurement." *MICCAI 2026* (Springer LNCS), published September 2026. Third author. doi:10.1007/978-3-032-38236-8_8

Kim, J., Jung, S., **Song, D. S.**, Lee, Y., Park, S., Do, J., Ro, D. H. "Image-Only Acquisition Triage for Orthopedic AI: Avoiding Unreliable DICOM Metadata and Circular Validation." *IEEE FMLDS 2026*, accepted (camera-ready version submitted). Third author.

Lee, D. W., **Song, D. S.**, et al. "Accurate, automated classification of radiographic knee osteoarthritis severity using a novel method of deep learning: Plug-in modules." *Knee Surgery & Related Research*, 2024. Second author; the paper's author statement lists my contribution as data acquisition and interpretation, while in practice I carried out the algorithm work end to end, from devising the method to implementation, training and experiments (named co-inventor on the related patent application). The paper reports 75.7% ensemble accuracy on an independent test set versus 62.0–72.8% for the individual models. doi:10.1186/s43019-024-00228-3

"Image size transformation data augmentation method for lesion segmentation on normalization and augmentation medical images." *Korea Software Congress (KSC)*, 2022, poster presentation. First author.

Patents & Other Research Output

- Inventor, Korean Patent Application 10-2024-0124529 and PCT/KR2025/012689: ensemble system for classifying knee osteoarthritis severity (the plug-in-module method above).
- Inventor, Korean Patent Application 10-2025-0043898: generating a patient's projected knee image from the current radiograph and its K–L/OARSI grades.

Technical Strengths

Research	Controlled ablation design, multi-seed evaluation, intervention-based functional analysis, representation analysis, reproducibility
ML	PyTorch, PyTorch Lightning, Hydra, multi-task and partial-label learning, foundation-model adaptation (linear probing, LoRA), medical-image classification/segmentation
Compression & serving	ONNX, OpenVINO 16-bit, quantization and structured/block pruning, TorchServe, BentoML, FastAPI, vLLM
Engineering	Python, C/C++, Docker, GPU server operation (H100, L40S), regulated-product documentation

Languages

Korean (native); English (conversational); Japanese (conversational).